Vance_week_5_6_assignment

2025-10-19

```
# IMPORTS

library(dplyr)
```

```
##
## Attaching package: 'dplyr'
```

```
## The following objects are masked from 'package:stats':
##
##     filter, lag
```

```
## The following objects are masked from 'package:base':
##
##     intersect, setdiff, setequal, union
```

```
library(ggplot2)
library(readxl)
library(tidyverse)
```

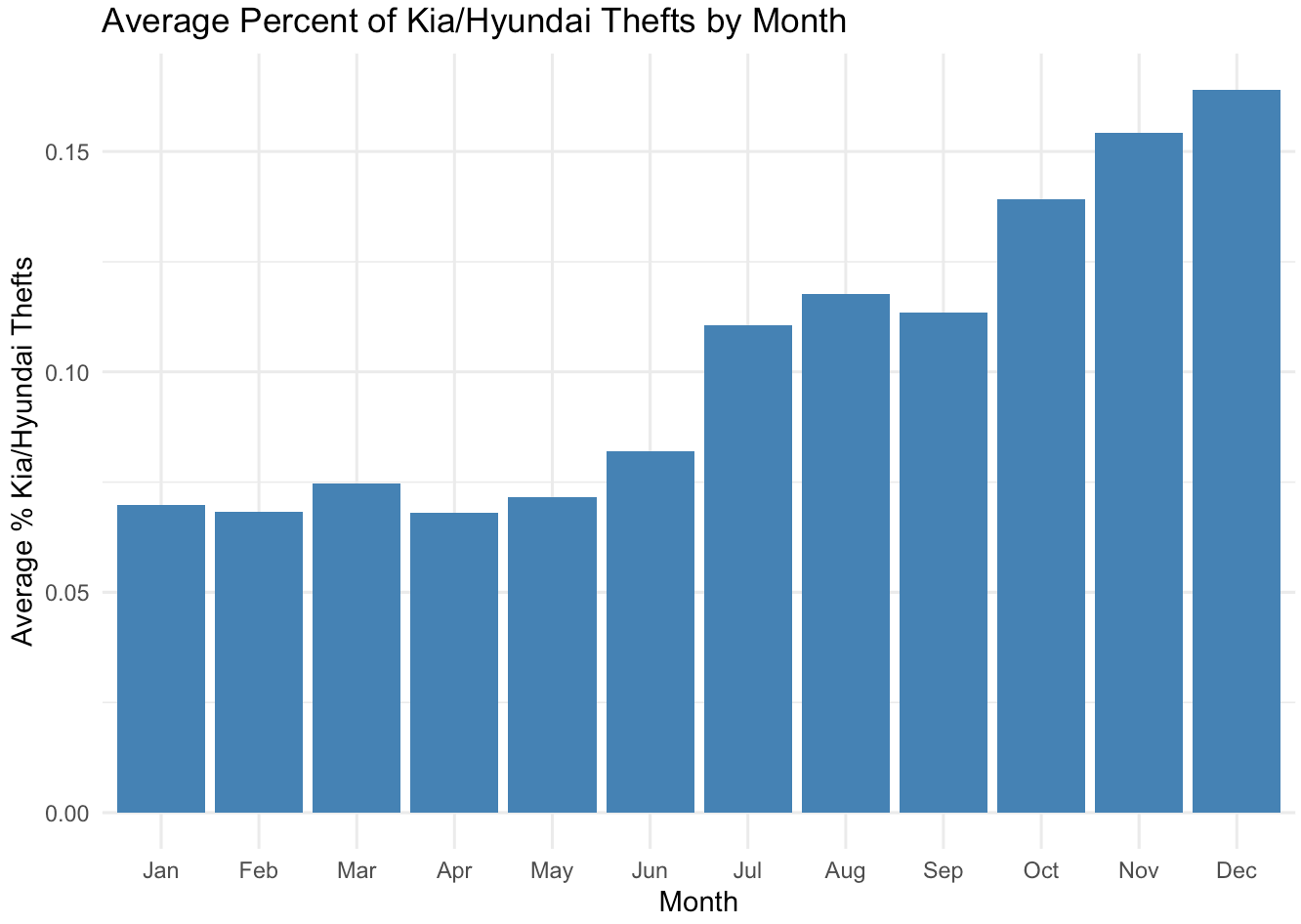
```
## ── Attaching core tidyverse packages ──────────────────────── tidyverse 2.0.0 ──
## ✔ forcats   1.0.0     ✔ stringr   1.5.1
## ✔ lubridate 1.9.4     ✔ tibble    3.2.1
## ✔ purrr     1.0.4     ✔ tidyr     1.3.1
## ✔ readr     2.1.5
```

```
## ── Conflicts ────────────────────────────────────────── tidyverse_conflicts() ──
## ✖ dplyr::filter() masks stats::filter()
## ✖ dplyr::lag()    masks stats::lag()
## ℹ Use the conflicted package (<http://conflicted.r-lib.org/>) to force all conflicts
to become errors
```

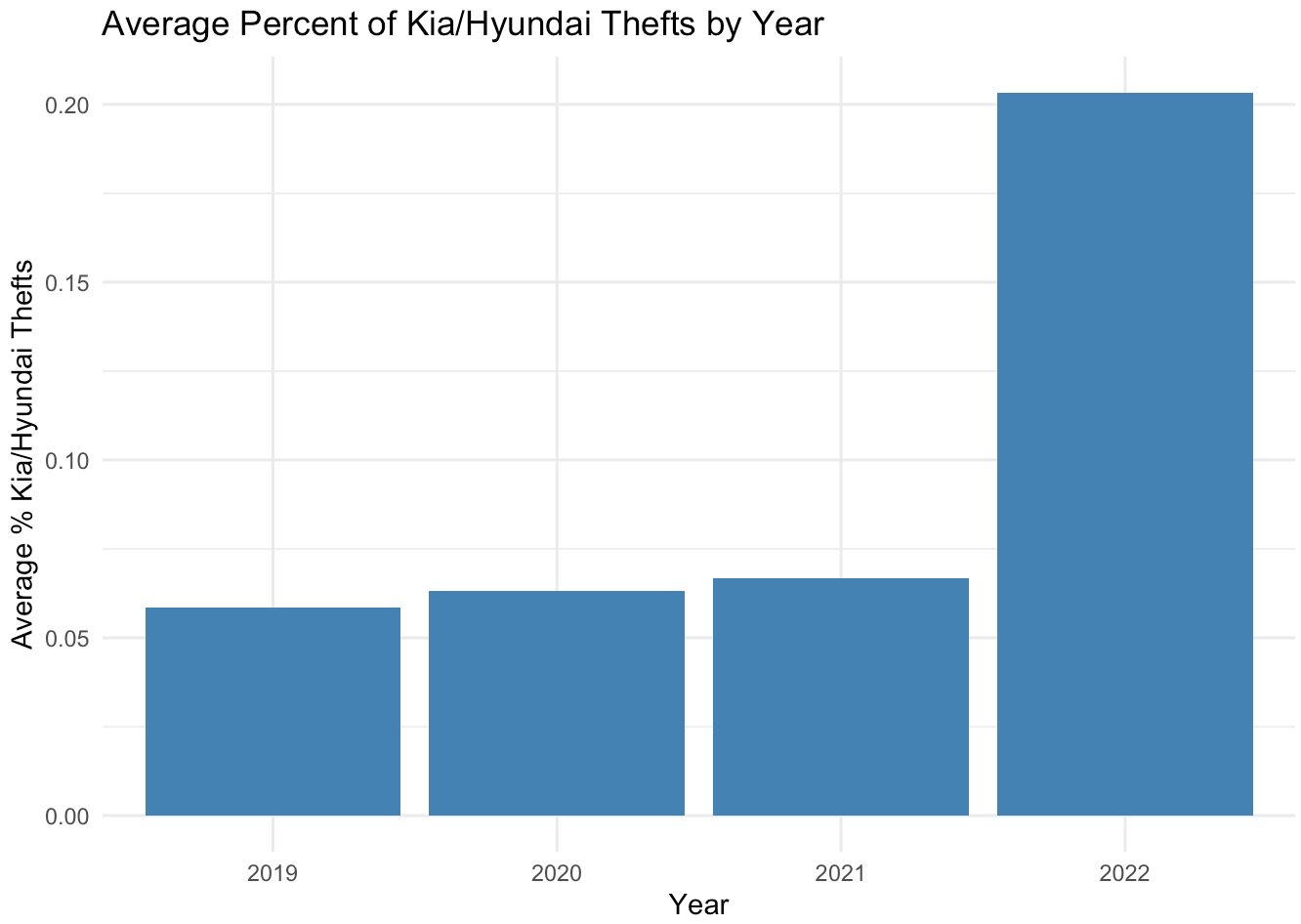
```
library(treemapify)
```

```
kiaHyundaiThefts <- read.csv("kiaHyundaiThefts.csv")

# print the shape
kiaHyundaiThefts %>%
  group_by(month) %>%
  summarise(across(starts_with("count"), mean),
            percentKiaHyundai = mean(percentKiaHyundai)) %>%
  mutate(month = factor(month, levels = month.abb)) %>%
  ggplot(aes(x = month, y = percentKiaHyundai)) +
  geom_col(fill = "steelblue") +
  labs(title = "Average Percent of Kia/Hyundai Thefts by Month",
       x = "Month", y = "Average % Kia/Hyundai Thefts") +
  theme_minimal()
```




```
kiaHyundaiThefts %>%
  group_by(year) %>%
  summarise(across(starts_with("count"), mean),
            percentKiaHyundai = mean(percentKiaHyundai)) %>%
  mutate(year = factor(year)) %>%
  ggplot(aes(x = year, y = percentKiaHyundai)) +
  geom_col(fill = "steelblue") +
  labs(title = "Average Percent of Kia/Hyundai Thefts by Year",
       x = "Year", y = "Average % Kia/Hyundai Thefts") +
  theme_minimal()
```

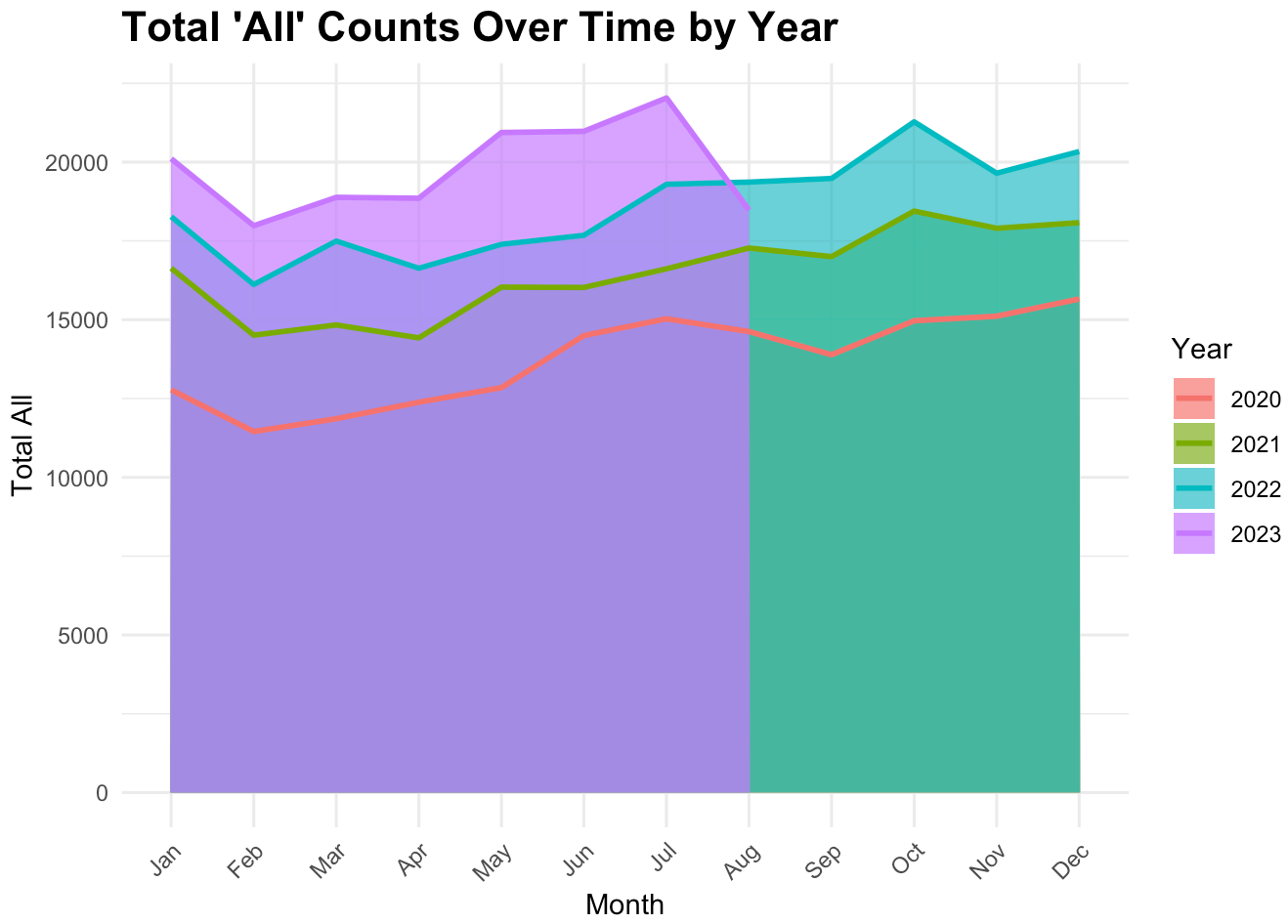

Average Percent of Kia/Hyundai Thefts by Year
Average % Kia/Hyundai Thefts
0.20
0.15
0.10
0.05
0.00
2019
2020
2021
2022
Year

```
df <- read.csv("tidy_data.csv")
# --- Clean and prepare data ---

df <- df %>%
  # Convert Date to Date type and extract Year and Month
  mutate(
    Date = as.Date(Date),
    Year = year(Date),
    Month = month(Date, label = TRUE, abbr = TRUE)
  )

# --- Summarize total "All" by Year-Month ---
df_summary <- df %>%
  # 2019 is incomplete, so we'll exclude it
  filter(Year != 2019) %>%
  # Aggregate totals
  group_by(Year, Month) %>%
  summarize(TotalAll = sum(All, na.rm = TRUE), .groups = "drop")

# --- Plot area chart ---
ggplot(df_summary, aes(x = Month, y = TotalAll, fill = factor(Year), group = Year)) +
  geom_area(alpha = 0.7, position = "identity") + # Stacked area
  geom_line(aes(color = factor(Year)), linewidth = 1) +
  # Customize labels and theme
  labs(
    title = "Total 'All' Counts Over Time by Year",
    x = "Month",
    y = "Total All",
    fill = "Year",
    color = "Year"
  ) +
  theme_minimal() +
  # Customize theme
  theme(
    plot.title = element_text(size = 16, face = "bold"),
    axis.text.x = element_text(angle = 45, hjust = 1)
  )
```


Total 'All' Counts Over Time by Year
Total All
20000
15000
10000
5000
0
Jan
Feb
Mar
Apr
May
Jun
Jul
Aug
Sep
Oct
Nov
Dec
Month
Year
2020
2021
2022
2023

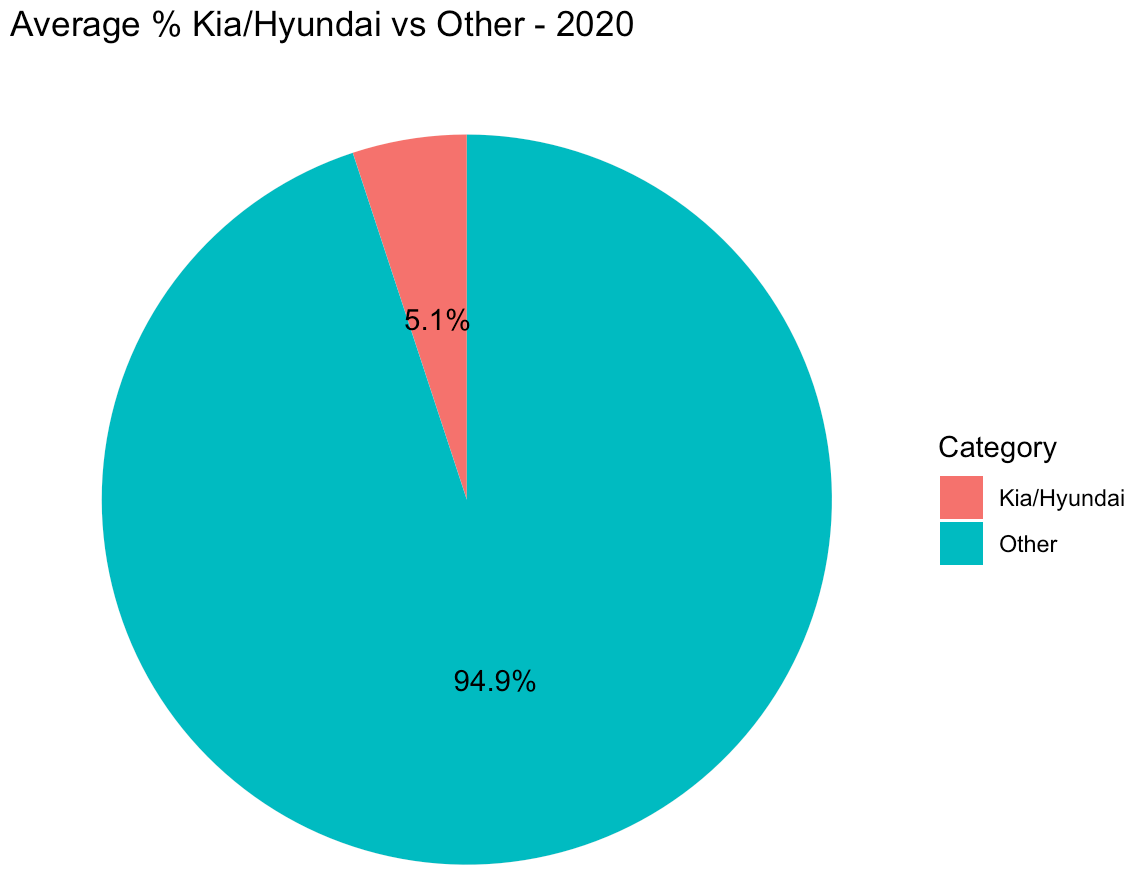
```
# Extract year and convert Percent to numeric
df <- df %>%
  mutate(year = substr(Date, 1, 4),
         Percent = as.numeric(Percent))

# Function to plot pie chart for a given year
plot_pie_year <- function(df, year_val) {
  avg_percent <- df %>%
    filter(year == year_val) %>%
    summarise(mean_percent = mean(Percent, na.rm = TRUE)) %>%
    pull(mean_percent)

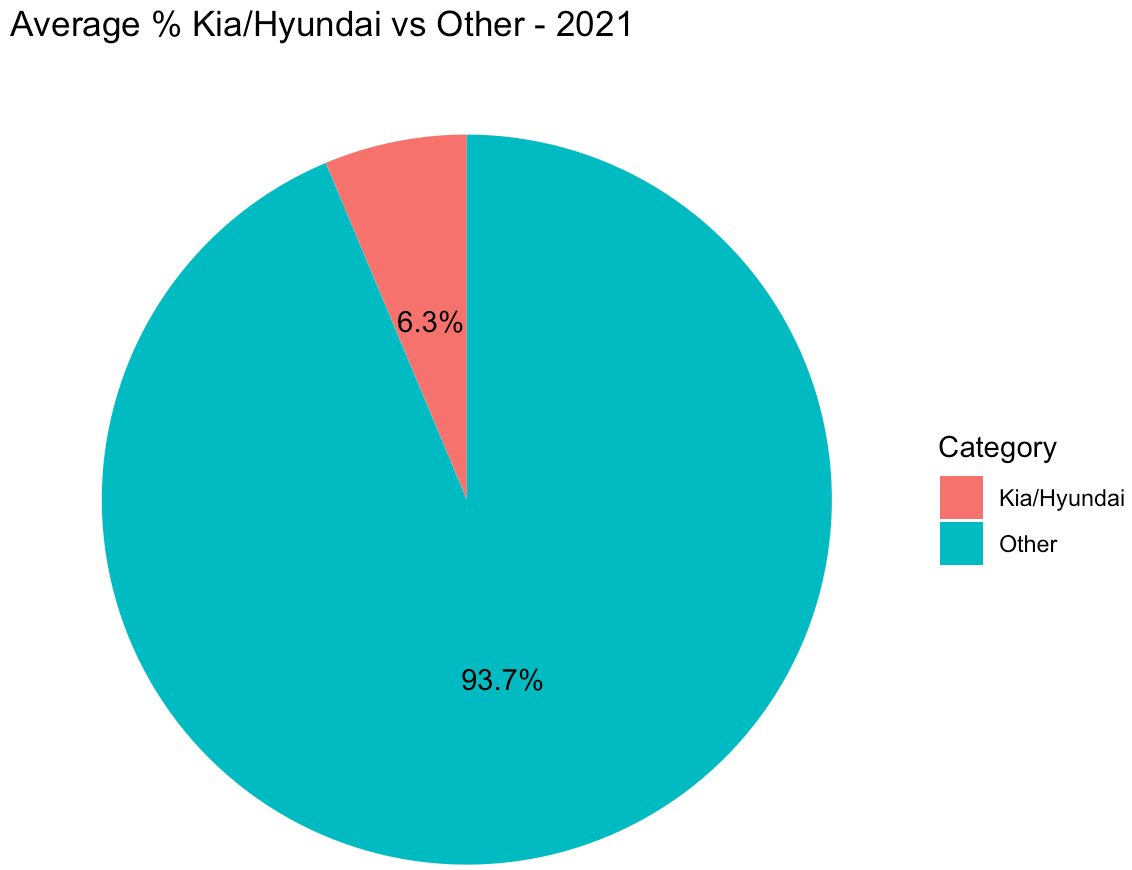
  pie_df <- data.frame(
    Category = c("Kia/Hyundai", "Other"),
    Percent = c(avg_percent, 100 - avg_percent)
  )

  ggplot(pie_df, aes(x = "", y = Percent, fill = Category)) +
    geom_bar(stat = "identity", width = 1) +
    coord_polar("y") +
    labs(title = paste("Average % Kia/Hyundai vs Other -", year_val),
         fill = "Category") +
    theme_void() +
    geom_text(aes(label = paste0(round(Percent, 1), "%")),
              position = position_stack(vjust = 0.5))
}

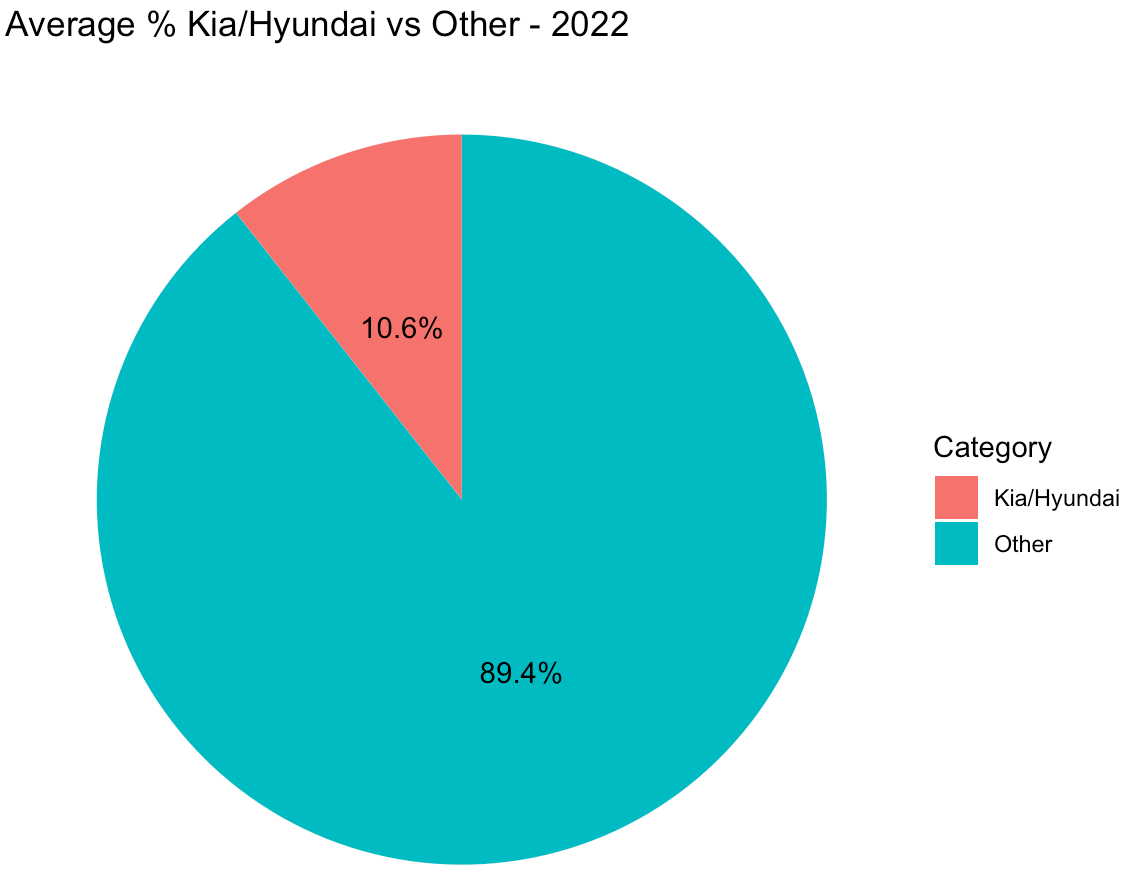
# Plot for 2020
plot_pie_year(df, "2020")
```




```
# Plot for 2021
plot_pie_year(df, "2021")
```




```
# Plot for 2022
plot_pie_year(df, "2022")
```

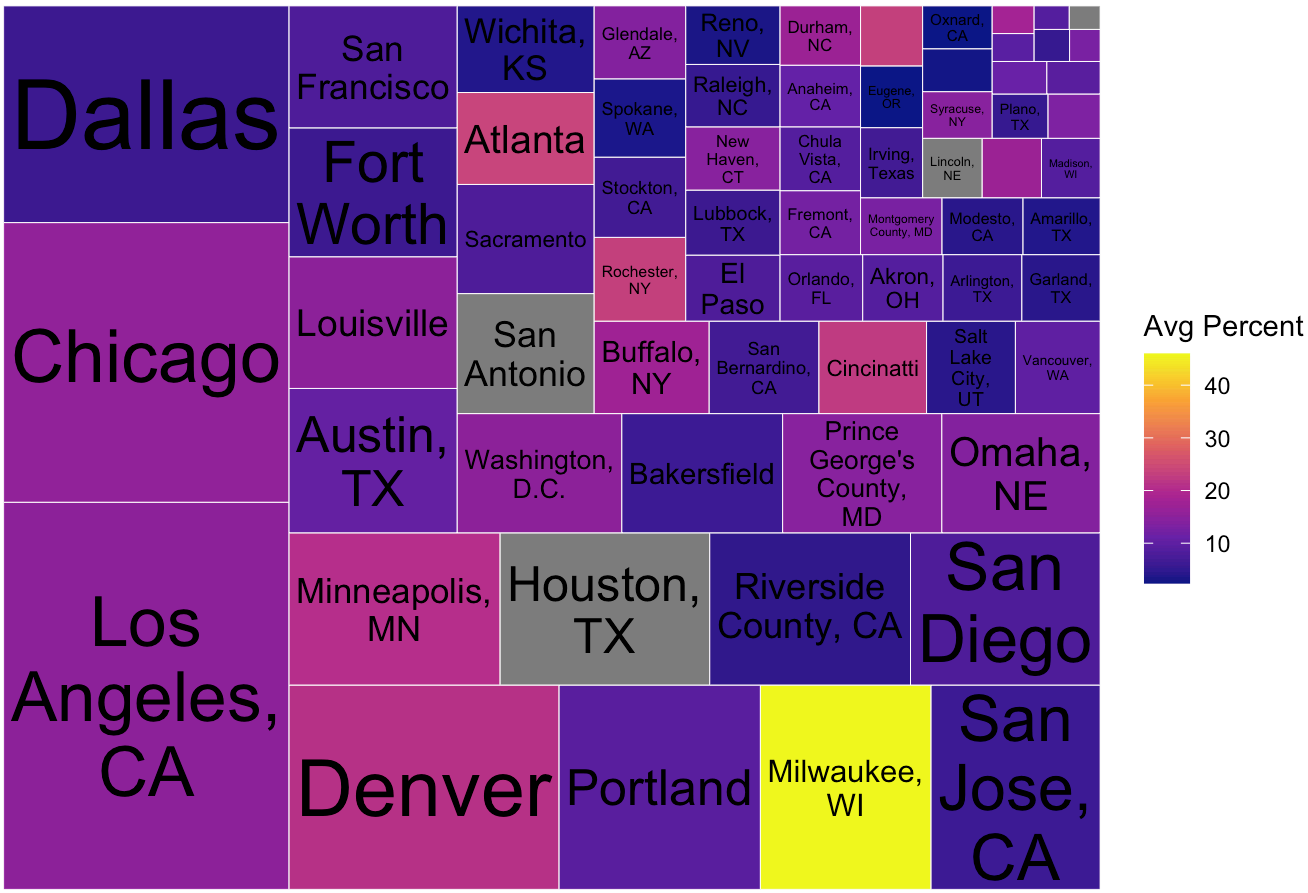

Average % Kia/Hyundai vs Other - 2022
10.6%
89.4%
Category
Kia/Hyundai
Other

```
# --- Clean and prepare data ---
df <- df %>%
  mutate(
    Date = as.Date(Date),
    Year = lubridate::year(Date)
  )

# --- Aggregate totals by Location (optional if multiple dates) ---
df_summary <- df %>%
  group_by(Location) %>%
  summarize(
    TotalAll = sum(All, na.rm = TRUE),
    AvgPercent = mean(Percent, na.rm = TRUE)
  )

# --- Treemap ---
ggplot(df_summary, aes(
  area = TotalAll,
  fill = AvgPercent,
  label = Location
)) +
  geom_treemap(color = "white") +
  geom_treemap_text(
    colour = "black",
    place = "centre",
    grow = TRUE,
    reflow = TRUE
  ) +
  scale_fill_viridis_c(option = "plasma") +
  labs(
    title = "Vehicle Counts by Location (Treemap)",
    fill = "Avg Percent"
  ) +
  theme_minimal()
```

Vehicle Counts by Location (Treemap)



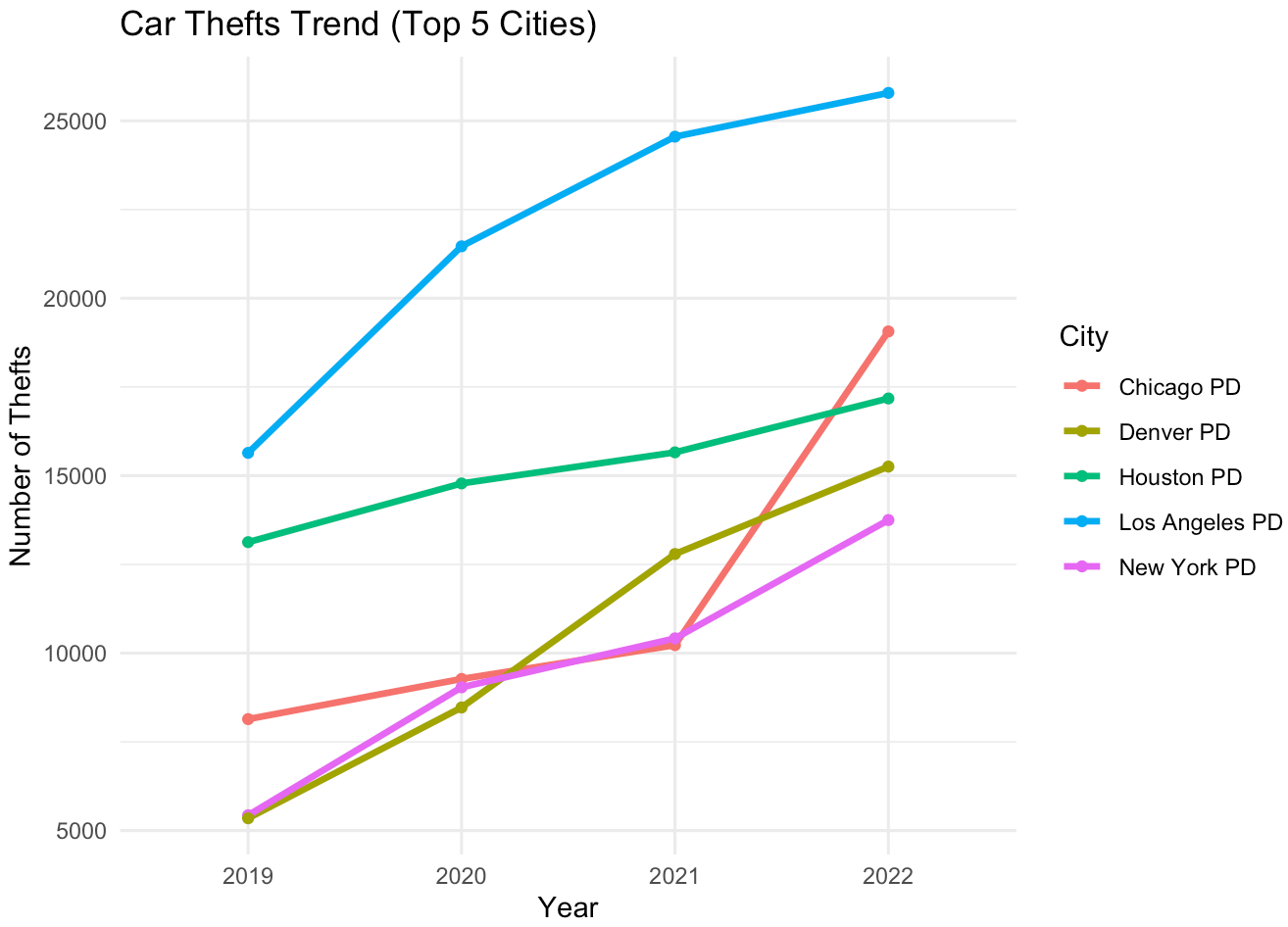
```
df <- read.csv("carTheftsMap.csv", stringsAsFactors = FALSE)

# Convert "-" to NA and columns to numeric
year_cols <- c("countCarThefts2019", "countCarThefts2020", "countCarThefts2021", "countC
arThefts2022")
df[year_cols] <- lapply(df[year_cols], function(x) as.numeric(gsub("-", NA, x)))

# Gather into long format
df_long <- df %>%
  select(geo_name, all_of(year_cols)) %>%
  pivot_longer(-geo_name, names_to = "Year", values_to = "Thefts") %>%
  mutate(Year = gsub("countCarThefts", "", Year))

# Example: Top 5 cities by 2022 thefts
top_cities <- df %>%
  arrange(desc(countCarThefts2022)) %>%
  slice_head(n = 5) %>%
  pull(geo_name)

ggplot(df_long %>% filter(geo_name %in% top_cities), aes(x = Year, y = Thefts, color = g
eo_name, group = geo_name)) +
  geom_line(size = 1.2) +
  geom_point() +
  labs(title = "Car Thefts Trend (Top 5 Cities)", x = "Year", y = "Number of Thefts", co
lor = "City") +
  theme_minimal()
```



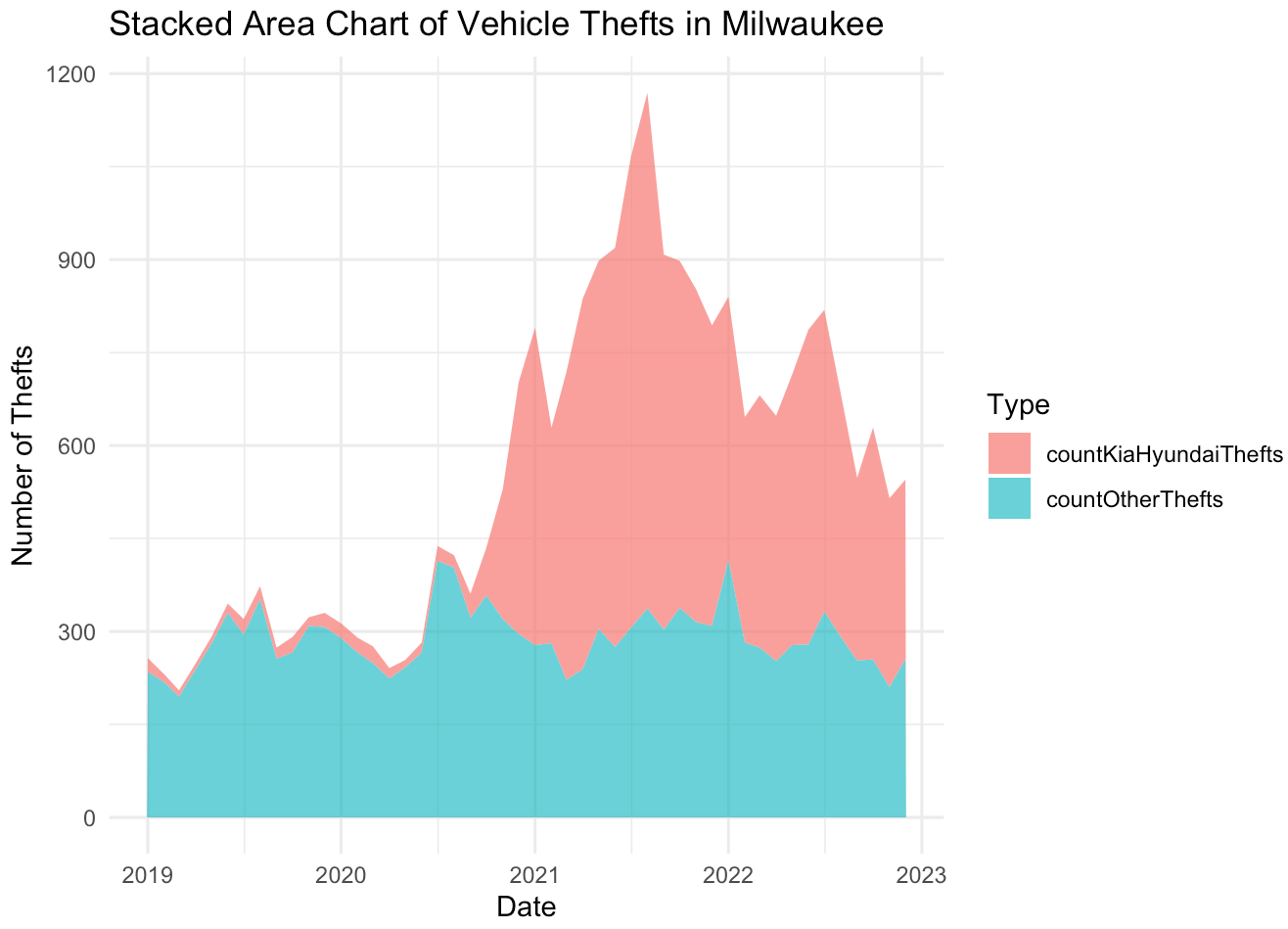

```
# Read the data
df <- read_csv("KiaHyundaiMilwaukeeData.csv")
```

```
## Rows: 48 Columns: 7
## ── Column specification ────────────────────────────────────────────────────────
## Delimiter: ","
## chr (3): month, city, state
## dbl (4): year, countKiaHyundaiThefts, countOtherThefts, percentKiaHyundai
##
## ℹ Use `spec()` to retrieve the full column specification for this data.
## ℹ Specify the column types or set `show_col_types = FALSE` to quiet this message.
```

```
# Create a date column for proper ordering
df <- df %>%
  mutate(date = as.Date(paste(year, month, "01", sep = "-"), format = "%Y-%b-%d"))

# Reshape to long format for stacking
df_long <- df %>%
  select(date, countKiaHyundaiThefts, countOtherThefts) %>%
  pivot_longer(cols = c(countKiaHyundaiThefts, countOtherThefts),
               names_to = "Type", values_to = "Count")

# Plot stacked area chart
ggplot(df_long, aes(x = date, y = Count, fill = Type)) +
  geom_area(alpha = 0.7) +
  labs(title = "Stacked Area Chart of Vehicle Thefts in Milwaukee",
       x = "Date", y = "Number of Thefts", fill = "Type") +
  theme_minimal()
```



Call to Action

The current administration has recognized the alarming rates of car thefts. They realize thefts vary by location and time, but the public must be made aware of the crime rates. Our call to action is to inform the general public of crime trends so they can take precautions if they live in an area witnessing an increase in crime.

Audience

General Public

Good evening. My name is Jameson Rodgers, data analyst for the FBI. Last month we were tasked with reporting the overall crime trends in the USA as it pertains to car theft. Utilizing multiple data sources, we saw the same trend: car thefts have increased from 2019 to 2022. As you can tell from this chart, 2019 is denoted with the lowest cumulative car thefts, and each year surpasses the previous.

We examined the top cities and saw similar trends. Despite being in different regions, Chicago, Denver, Houston, Los Angeles, and New York all saw increases in car thefts from 2019 to 2022.

Additionally, we examined Hyundai and Kia thefts across the country. Most areas only saw these cars make up a small percentage of total car thefts. However, in Milwaukee, these two brands made up almost half of all car thefts.

In conclusion, we want to provide an alert to the public: car thefts are up. Your car is at at risk of being stolen. We are actively working with law enforcement to deter car thefts and reverse this trend. To the citizens of Milwaukee, we recognize an uptick in thefts for Kias and Hyundais. This is due to a manufacturing error that makes these cars easier to steal. Please consult your local body shop to fix this flaw. The attorney general will be seeking charges against Hyundai and Kia for negligence in this matter. Thank you for your time.